CS412 Project on Earthquake Prediction using Phonetic Data

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1 Introduction

Reliable earthquake forecasting remains an unresolved problem in seismology, and improved prediction could help reduce damage to infrastructure and human life. This project studies whether useful indicators of future failure are hidden in laboratory acoustic seismic signals. We hypothesize that stress buildup before failure may produce changes in waveform regularity, intensity, and frequency composition, analogous to how prosodic structure in speech reflects changes in vocal dynamics. Motivated by this analogy, we treat seismic acoustic emissions as non-stationary one-dimensional signals and extract speech-inspired acoustic features to predict the remaining time before the next failure event.

Specifically, we compare two feature extraction pipelines: handcrafted eGeMAPS features Jadoul et al. [2018], Boersma and Weenink [2021], Eyben et al. [2010] and data-driven MiniRocket features Dempster et al. [2021]. The eGeMAPS pipeline summarizes interpretable acoustic properties such as formant structure, loudness, perturbation, harmonicity, spectral shape, and cepstral structure, while MiniRocket automatically extracts high-dimensional convolutional features from raw waveform segments. These representations are evaluated with several regression models, including LightGBM Ke et al. [2017], XGBoost Chen et al. [2015], Random Forest Breiman [2001], and SVR Drucker et al. [1996], using both Kaggle private test MAE and five-fold cross-validation MAE.

Our results show that: i) LightGBM achieves the best overall performance among the tested baselines, outperforming XGBoost, Random Forest, and SVR. ii) Feature analysis further reveals that the top eGeMAPS feature and the top MiniRocket feature are highly correlated, suggesting that the handcrafted and data-driven pipelines independently capture a similar predictive signal pattern. iii) Finally, the preprocessing ablation shows that wavelet denoising improves eGeMAPS features but hurts MiniRocket features. This indicates that preprocessing should be matched to the feature extractor: handcrafted acoustic features benefit from noise suppression because they rely on stable numerical waveform properties, whereas MiniRocket may exploit fine-grained temporal textures that aggressive denoising can remove.

2 Related Work

The three winning Kaggle solutions for laboratory earthquake prediction reveal several effective feature and model choices. For feature engineering, peak-related statistics, such as peak counts at different height thresholds and on denoised signals, were important across multiple solutions. Robust distributional features, including percentile differences (e.g., 95%–5% and 80%–20%), reduce dependence on mean drift and provide stable summaries of signal variability. Frequency-domain descriptors, such as Mel-frequency cepstral coefficients (MFCCs) and binned Fast Fourier Transform (FFT) power spectra, also improve the ability to capture waveform patterns. The first-place solution used an important preprocessing step: adding small Gaussian noise to each segment and subtracting the segment median, which helps reduce time-related spurious correlations. Other useful features included rolling-window standard deviation percentiles, truncated kurtosis, and robust regression trend slopes.

For model selection, LightGBM Ke et al. [2017], a histogram-based gradient boosting method with leaf-wise tree growth, served as the primary model for two champion teams due to its efficiency and strong generalization. Further gains came from multi-task learning or blending with SVR Drucker et al. [1996] and neural networks. CatBoost Dorogush et al. [2018], trained on many randomly sampled segments, also achieved excellent scores. Recurrent neural networks such as GRU Cho et al. [2014] obtained respectable performance even with only a small number of features. Beyond the original competition, recent time-series methods such as MiniRocket Dempster et al. [2021] provide efficient automatic feature extraction using fixed random convolutional kernels. Overall, prior solutions suggest that peak-related features, robust distributional statistics, frequency-domain descriptors, and ensemble or hybrid models are especially effective for this task.

3 Methodology

3.1 Data preprocessing

Before feature extraction, we include an optional denoising step. We use wavelet denoising because it preserves both temporal and frequency-localized information, making it more suitable than Fourier-based filtering for signals with local fluctuations and transient structures. Specifically, the signal is decomposed using a Daubechies-4 wavelet, and a fixed hard threshold is applied to the resulting wavelet coefficients. Coefficients with absolute values below the threshold are set to zero, while larger coefficients are retained unchanged. The denoised signal is then reconstructed from the remained coefficients.

3.2 Feature extraction

eGeMAPS feature extraction. For each seismic waveform segment, we extracted a compact set of acoustic-inspired summary features using Parselmouth/Praat and OpenSMILE Jadoul et al. [2018], Boersma and Weenink [2021], Eyben et al. [2010]. These features were originally developed for speech analysis. In this task, the goal is to predict the remaining time to the next failure event; therefore, these features are used to capture signal changes that may correlate with the evolving stress state of the system. The final feature set contains 54 input features in total. First, we extracted 18 low-level descriptors (LLDs) using Parselmouth and summarized each descriptor over the waveform segment using two statistics: the mean and the coefficient of variation (CV), resulting in 36 features. The mean captures the average level of each acoustic property, while the coefficient of variations capture its relative temporal variability. These compose the minimalistic GeMAPS set. Second, we added 18 OpenSMILE extended GeMAPS (eGeMAPS) functionals Eyben et al. [2016], audEERING GmbH, including loudness percentiles, loudness rising/falling slopes, spectral flux statistics, and MFCC-based cepstral descriptors. These additional features capture distributional, dynamic, and cepstral properties. The extracted features can be grouped into several categories. (i) Fundamental-frequency features capture the dominant vibration pattern of the seismic waveform. (ii) Local perturbation features describe cycle-to-cycle irregularity in frequency and amplitude; in the seismic setting, increased irregularity may indicate more unstable or heterogeneous wave propagation before failure. (iii) Formant and resonance-related features characterize the main spectral peaks of the seismic signal, which may reflect how the fault system amplifies specific frequency components at different stages before failure. (iv) Energy and loudness features summarize the overall signal intensity and its temporal dynamics, including loudness distribution and rising/falling slopes; these features may be useful because seismic activity can become stronger or more burst-like as failure approaches. (v) Spectral-shape features describe whether the waveform is dominated by low-frequency, high-frequency, or broadband components; this frequency-balance information is important because the spectral content of seismic signals may evolve as the system ruptures. (vi) Spectral-dynamics features, such as spectral flux, measure the frame-to-frame rate of change in spectral content. Increasing spectral flux may indicate that the waveform becomes more temporally unstable, with rapid fluctuations in frequency composition as failure approaches. (vii) Formant-energy features quantify how strongly signal energy is concentrated around resonance-like peaks rather than broadly distributed across the spectrum, which may reflect changes in damping or resonance behavior of the subsurface medium. (viii) Cepstral and timbral features, such as MFCCs, encode the overall shape of the spectral envelope using a compact set of coefficients. In the seismic setting, they capture the gross timbral texture of the acoustic emission and may reflect changes in dominant propagation modes or wave types as stress accumulates. (ix) Harmonic-structure features describe waveform regularity and help distinguish structured seismic wave propagation from more chaotic vibration patterns before failure.

MiniRocket feature extraction. MiniRocket Dempster et al. [2021] is a strong and efficient algorithm specifically designed for time-series tasks. Unlike our proposed methodology, which relies on domain-inspired feature engineering, MiniRocket operates directly on the raw continuous acoustic data, automatically extracting features using a fixed set of random convolutional kernels. We include MiniRocket to serve as a robust time-series method baseline. By benchmarking our feature design against it, we can verify whether our handcrafted phonetic features can represent the underlying pre-earthquake mechanisms effectively.

3.3 Models

Support Vector Regression. To establish a strong, non-linear baseline, we employ Support Vector Regression Drucker et al. [1996]. Using a RBF kernel, SVR captures continuous, non-linear relationships between features and predictions. Crucially, SVR (including all the other baselines) was trained on the exact same handcrafted phonetic features as our proposed LightGBM model. This provides a direct, classical machine learning counterpart to our modern gradient boosting approach.

Random Forest. We also implement a Random Forest regressor Breiman [2001]. By constructing an ensemble of independent decision trees via bagging, Random Forest naturally mitigates the high variance and overfitting risks inherent in single decision trees.

XGBoost. Unlike the parallel bagging approachs, XGBoost Chen et al. [2015] builds decision trees sequentially, with each new tree designed to minimize the residual errors of the previous ensemble.

LightGBM. We train a modern gradient boosting decision tree model for this task because they are suitable for smaller, noisy feature sets with many values as in the case for our windowed acoustic data. Specifically, we chose LightGBM for its high accuracy and efficient training times.

All the training details are included in the next section.

4 Detailed Experiment Setup

The experiment pipeline follows diagram inside Figure1

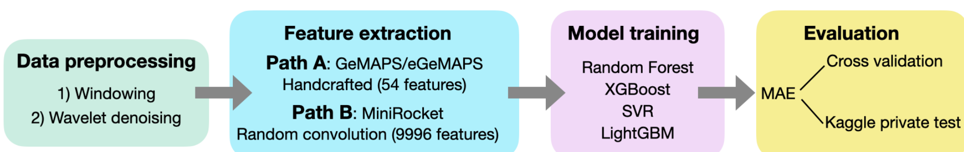

Figure 1: Experiment pipeline

4.1 Preprocessing

The raw training data is a continuous acoustic trajectory with 629,145,480 time steps. We split it into non-overlapping windows of 150,000 points, yielding 4,194 training segments. Each segment is labeled by the `time_to_failure` value at its final time step, representing the remaining time until the next earthquake event. The 2,624 test segments are provided pre-segmented. Before handcrafted feature extraction, each segment is denoised by db4 wavelet thresholding: coefficients with absolute value below 10.0 are set to zero before reconstruction. The reconstructed waveform is then used for eGeMAPS feature extraction.

4.2 Feature Extraction

For the eGeMAPS pipeline, we extract 54 acoustic-inspired features using Parselmouth/Praat and OpenSMILE. For the MiniRocket pipeline, each raw segment is passed directly into MiniRocket, which uses approximately 10,000 random convolutional kernels and summarizes each response by the proportion of positive values (PPV), producing a 9,996-dimensional input. Unlike eGeMAPS, MiniRocket does not require handcrafted domain-specific features.

4.3 Training and Evaluation

Random Forest uses absolute error as the split criterion, considers $\sqrt{d}$ candidate features per split, and trains 200 trees per fold. **SVR** uses `StandardScaler` before training because the RBF kernel is scale-sensitive, with $C = 10.0$ and $\epsilon = 0.1$. **XGBoost** uses absolute error as the objective, with 500 trees, learning rate 0.05, maximum depth 6, and 0.8 row/column subsampling. **LightGBM** is trained with Fair loss as a smooth MAE approximation. For eGeMAPS features, we use learning rate 0.02, 16 leaves, 0.8 feature/bagging fractions, and 0.1 L1/L2 regularization, with early stopping. For MiniRocket features, we use learning rate 0.05, 31 leaves, and feature fraction 0.1 to reduce overfitting and computation on the high-dimensional input. We evaluate performance using two MAE metrics. Kaggle private test MAE is obtained from submissions to the hidden test set, whose labels are inaccessible. Cross-validation MAE is computed using five shuffled folds over all available training data with random seed 42 and reported as mean $\pm$ standard deviation. For Kaggle test prediction, outputs are averaged across the five fold-specific models.

4.4 Feature Analysis

For tree-based models, we use *gain* to quantify feature importance by measuring how much each feature improves the model objective when it is used for a split. In boosted decision trees, the gain of a feature is accumulated over all splits involving that feature, so features with larger total gain contribute more to reducing prediction error. Compared with split-count importance, gain is more informative because it reflects the quality of the splits rather than only how often a feature is selected XGBoost Developers.

5 Results

5.1 Model Selection Results

Table 1 reports the performance of four models trained on wavelet-denoised eGeMAPS features. LightGBM achieves the best performance on both the Kaggle private test set and cross-validation, while SVR performs worst on both metrics.

Table 1: Model comparison, feature ablation, and preprocessing ablation results.

Category	**Setting**	**Kaggle MAE (Private) ↓**	**CV MAE ↓**	**CV RMSE ↓**
Different models	XGBoost	2.56951	$2.0821^{\pm 0.0393}$	$2.6447^{\pm 0.0507}$
	Random Forest	2.52600	$2.0851^{\pm 0.0521}$	$2.6732^{\pm 0.0654}$
	SVR	2.64636	$2.1450^{\pm 0.0954}$	$2.7952^{\pm 0.0669}$
	LightGBM	**2.50571**	$\mathbf{2.0368^{\pm 0.0387}}$	$\mathbf{2.6414^{\pm 0.0487}}$
Feature ablation	MiniRocket	2.57144	$2.0937^{\pm 0.0424}$	$2.6687^{\pm 0.0474}$
	eGeMAPS	**2.50571**	$\mathbf{2.0368^{\pm 0.0387}}$	$\mathbf{2.6414^{\pm 0.0487}}$
Preprocessing ablation	**With wavelet denoising**	**2.50571**	$\mathbf{2.0368^{\pm 0.0387}}$	$\mathbf{2.6414^{\pm 0.0487}}$
	w/o wavelet denoising	2.59688	$2.0764^{\pm 0.0472}$	$2.6884^{\pm 0.0589}$

LightGBM's advantage comes from two complementary factors. First, it uses sequential gradient boosting: each tree is trained to correct the residual errors of the previous ensemble, allowing the model to progressively capture complex, nonlinear interactions among 54 features. Second, we use the Fair loss objective, a smooth approximation to MAE that is less sensitive to high-amplitude transient bursts than squared error. Because seismic data contain outlier segments that would disproportionately penalize squared-error models, Fair loss leads to a more robust fit. Early stopping further prevents overfitting by stopping training once validation MAE stops improving. The low cross-validation standard deviation suggests that LightGBM generalizes stably across folds. Random Forest and XGBoost are competitive but fall short. Random Forest builds trees independently and averages their predictions, which reduces variance but cannot iteratively correct errors because it lacks the refinement mechanism of boosting. XGBoost's level-wise tree growth is less efficient at finding high-impact splits than LightGBM's leaf-wise strategy. SVR underperforms for several reasons. Structurally, SVR is a single model without ensembling, making it more sensitive to variance. Its RBF kernel maps features into an implicit high-dimensional space and optimizes a margin-based objective, which does not directly minimize MAE and may not align well with the regression task. Also, SVR has the largest cross-validation standard deviation (0.0954), more than twice that of LightGBM, indicating that its performance is sensitive to which segments fall into the training and validation folds. This instability likely contributes to poor generalization on the held-out test set. Interestingly, while LightGBM's cross-validation RMSE still outperforms the other model, it's improvement in this metric is less profound. This implies there are a substantial amount of large errors pulling RMSE upwards compared to the other models, considering the contrast with MAE.

5.2 Feature Extraction Results

As shown in Table 1, eGeMAPS features achieve a better performance than MiniRocket features. We attribute this to domain alignment: eGeMAPS are designed to capture physically interpretable properties, which have direct analogues in seismic waveforms. In contrast, MiniRocket applies randomly weighted convolutional kernels without domain knowledge and summarizes each response as a single scalar. With only 4,194 training segments, a high-dimensional representation is more susceptible to overfitting.

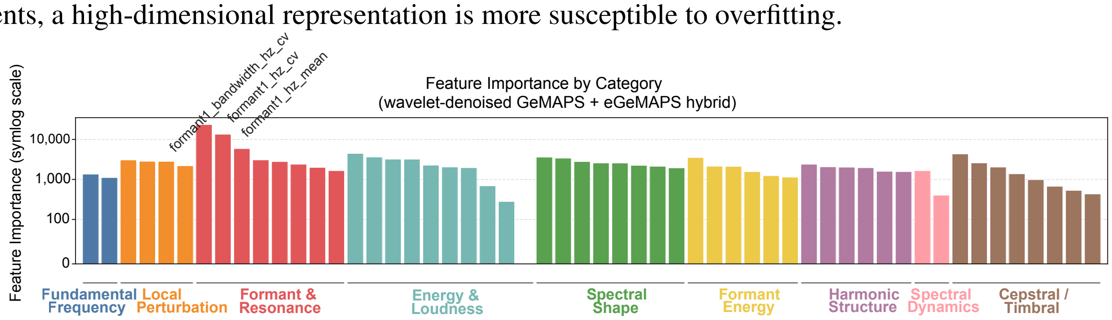

Figure 2: Feature importance

Examining the LightGBM gain importance for the eGeMAPS pipeline (Figure 2), the three highest-ranked features are all related to the first formant (F1): `formant1_bandwidth_hz_cv`, `formant1_hz_cv`, and `formant1_hz_mean`. F1 represents the lowest and most prominent spectral resonance, which may reflect bulk elastic properties of the rock surrounding the fault. As stress accumulates, micro-fracturing can alter effective stiffness and density, producing shifts in the dominant resonance frequency and greater frame-to-frame instability. These effects are captured by `formant1_hz_mean` and the CV-based F1 features, respectively. The dominance of F1 over higher formants is therefore physically reasonable, since F1 is more directly tied to bulk material changes, whereas higher formants describe finer structural details.

To examine whether the two pipelines capture overlapping or complementary information, we compute Pearson and Spearman correlations between selected features from each pipeline across all 4,194 training segments (Figure 3A). The top-ranked MiniRocket feature (`idx=3467`) shows near-perfect monotonic agreement with the highest-ranked eGeMAPS feature, (Figure 3A,B). MiniRocket feature `idx=3467` is a PPV statistic derived from a randomly weighted convolutional kernel applied at a fixed dilation. Its high correlation with F1 features suggests that this kernel's implicit frequency response is sensitive to energy in the F1 band. The two independent computational pathways: one grounded in acoustic phonetics theory and one data-driven, converge on the same underlying physical signal property. LightGBM's gain metric measures the total reduction in prediction loss attributable to splits on each feature. Once `formant1_bandwidth_hz_cv` has been exploited by early splits and captured most of the predictive variance, strongly correlated features contribute little additional gain because the data have already been partitioned along that dimension. Consequently, several MiniRocket features with zero gain importance still show high Spearman correlations above 0.93 with the top eGeMAPS features (Figure 3A). Low gain importance therefore does not necessarily mean that a feature is uninformative in absolute terms; rather, it may indicate redundancy with features already used by the model. The main exception is `loudness_sma3_percentile20.0` (Figure 3C), which shows near-zero correlation with all MiniRocket features ($|\rho| < 0.22$) and zero gain importance, more likely a floor-effect artifact rather than a redundant but informative signal.

5.3 Preprocessing Ablation Results

Table 1 reports the effect of wavelet denoising on the eGeMAPS feature pipeline. Applying db4 wavelet denoising with hard thresholding (threshold = 10.0) before feature extraction consistently improves both Kaggle private test MAE and cross-validation MAE. In contrast, denoising is detrimental for MiniRocket: the model trained on raw segments achieves a Kaggle MAE of 2.52068 and cross-validation MAE of

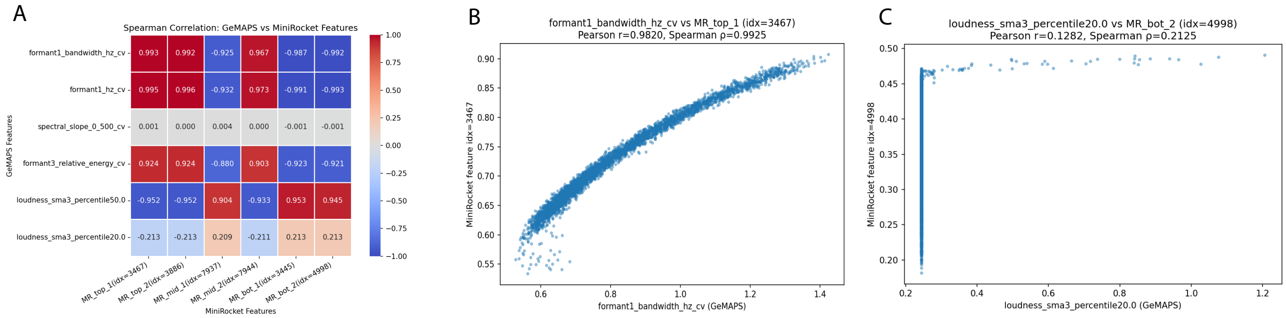

Figure 3: Feature correlation analysis between eGeMAPS and MiniRocket features. (A) Spearman correlation heatmap between selected eGeMAPS and MiniRocket features (top 2, middle 2, and bottom 2 by LightGBM gain importance). Color indicates correlation from -1 (blue) to $+1$ (red). (B) Scatter plot of the top-ranked eGeMAPS feature, `formant1_bandwidth_hz_cv`, and top-ranked MiniRocket feature, `idx=3467`, showing near-perfect monotonic agreement (Spearman $\rho > 0.99$). (C) The lowest-ranked eGeMAPS feature, `loudness_sma3_percentile20.0`, compared with the lowest-ranked MiniRocket feature, `idx=4998`; both show near-zero variance and negligible correlation.

2.0734 ± 0.0361 (not shown in the table for compactness), both better than the denoised variant (2.57144 and 2.0937 ± 0.0424). This asymmetry suggests that the two pipelines interact with preprocessing in fundamentally different ways. For eGeMAPS, wavelet denoising is beneficial because the handcrafted features are computed from explicit waveform properties, such as formant frequencies, spectral slopes, jitter, and shimmer, which are sensitive to high-frequency noise. Uncontrolled noise can introduce spurious spectral peaks and inflate perturbation-based measures. By zeroing out small wavelet coefficients, denoising removes these confounding components and allows the feature extractor to estimate resonance and spectral characteristics more reliably. The reduced cross-validation standard deviation (0.0287 vs. 0.0472) further suggests that denoising produces more consistent features across segments and stabilizes training. For MiniRocket, the same denoising step is harmful because MiniRocket does not measure predefined physical quantities. Instead, it applies thousands of random convolutional kernels at multiple dilations and converts each response into a PPV statistic, making it sensitive to local waveform motifs and fine-scale temporal textures. These small fluctuations may appear as noise to a handcrafted feature extractor, which averages or summarizes them over a segment, but they can still be useful to MiniRocket because different kernels can select and amplify recurring micro-patterns associated with stress evolution. Hard thresholding may therefore remove both noise and informative small-amplitude coefficients, smoothing the raw signal and reducing the diversity of temporal patterns available to the kernels. Overall, denoising helps feature extractors that rely on stable numerical waveform properties, but can hurt methods that exploit raw fine-grained temporal structure.

5.4 Future Work

The preprocessing ablation shows that different feature extractors benefit from different signal treatments: wavelet denoising improves eGeMAPS but hurts MiniRocket. This suggests that combining complementary feature extraction methods, possibly with feature-specific preprocessing, may further improve performance. In addition, top Kaggle solutions often rely on ensembles of different models. Although separating models is preferable in a research setting for clearer interpretability, practical prediction performance may benefit from ensembling LightGBM, XGBoost, Random Forest, SVR, and different feature representations.

Contributions Y. Z. implemented LightGBM, SVR, MiniRocket feature extraction and wavelet denoise.

J. L. implemented RandomForest and XGBoost, feature analysis for MiniRocket and eGeMAPS, wrote the final report.

B. C. designed and implemented eGeMAPS feature extraction, conducted Bayesian optimization hyperparameter searches (no significant difference from defaults, removed from main results considering compactness)

S. P. calculated RMSE cross validation values, added to discussion/edited reports.

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